

Arunkarthik Periyaswamy

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Skills

Languages: Java, Python, SQL, C++

Backend: Spring Boot, REST APIs, Microservices

Messaging: NATS JetStream, Kafka

Databases: MongoDB, MySQL, Oracle, Redis

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, Jenkins

Testing & Monitoring: JUnit, Mockito, Grafana, Postman

Professional Experience

ACI Worldwide

Jul 2025 - Present

Software Engineer

Remote, United States

- Built event-driven payment microservices with Java, Spring Boot, REST APIs, and NATS JetStream to support resilient, high-volume payment processing.
- Designed retry mechanisms and Circuit Breaker patterns across 12+ payment microservices, reducing transient failures and improving payment reliability.
- Developed automated connectivity monitoring for Fedwire, FedNow, CHIPS, and TCH RTP through administrative message exchanges, accelerating payment network failure detection and ensuring gateway availability.
- Built a streaming reporting engine with MongoDB cursors, batch processing, and JasperReports to generate 10GB+ transaction reports without memory bottlenecks.
- Implemented MongoDB TTL policies for administrative message logs, reducing storage utilization by 35% while meeting audit and compliance requirements.
- Resolved 65+ critical production issues across distributed services, messaging, and payment processing, improving platform stability and system availability.

Medtronic Labs

Dec 2024 - May 2025

ML/AI Research Intern

Remote, United States

- Evaluated open-source and proprietary speech AI models, selected a Whisper-based architecture, and drove end-to-end development of a multilingual healthcare conversational AI application.
- Built a conversational AI platform using Whisper, PyTorch, LLMs, and AWS SageMaker, reducing clinician documentation workload by 40%.
- Collaborated with cross-functional teams to integrate the solution into an existing healthcare platform and presented production-ready demos to engineering leadership.

Ideas2it Technologies Private Limited

Jan 2023 - Jan 2024

Senior Software Engineer

Chennai, India

- Ranked **50K+** resumes using uplift modeling with LangChain to prioritize candidates based on interview potential.
- Developed a React-based admin interface with TypeScript facilitating secure configuration of healthcare user operations.
- Transitioned a legacy Node.js system to a scalable Java-based microservices architecture with Spring Boot, enhancing system modularity, scalability, and deployment efficiency.
- Mentored and onboarded 2 interns, accelerating their project readiness to within one week

Ideas2it Technologies Private Limited

Jan 2021 - Jan 2023

Software Engineer

Chennai, India

- Built an automated recruitment tool using React, Whisper voice API and Flask that enabled hiring 150+ professionals.
- Engineered a Node.js-based microservices backend for a healthcare platform using Express.js to manage patient data, clinical workflows, and administrative operations.
- Enhanced CI/CD pipelines with Jenkins, streamlining testing and automation steps cutting deployment times by 30%.
- Automated the process of updating Oracle databases using Shell script and written unit tests using JUnit and Mockito.

Education

University at Buffalo

Jan 2024 - May 2025

Master of Science in Computer Science - Data Science

Buffalo, United States

Anna University

Aug 2016 - Sept 2020

Bachelor of Electrical and Electronics Engineering

Chennai, India

Awards

- ACI Worldwide Excellence Award – Recognized for resolving 65+ critical production issues that accelerated application production readiness and contributed to approximately \$1.5M in incremental business revenue.